Exercise 1.1

We construct a directed graph $G = (V, E, c)$ that models the problem. For each travel request there is one vertex v_i and there is one starting and final vertex s, t. Therefore:

$$V = \{s, v_1, \dots v_n, t\}$$

For each travel request (t_i, c_i) there exist two edges starting in its vertex v_i. One models the decision not to buy a BahnCard and connects v_i with v_{i+1} and has costs c_i. Additionally, there is an edge from that request to the request that has the lowest time t_j with $t_j > t_i + T$. This models the decision to buy a BahnCard at t_i. The edge has the costs of each of the travel requests that will be discounted multiplied by β.

$$\begin{aligned} E = &\{(v_i, v_{i+1}) \mid i \in \{1, \dots, n\} \text{ with } c(v_i, v_{i+1}) = c_i\} \\ &\cup \{(v_i, v_j) \mid i \in \{1, \dots, n\} \text{ where } t_j \text{ is next request after } t_i + T \\ &\text{with } c(v_i, v_j) = \sum_{k: t_k \in [t_i, t_i + T)} \beta \cdot c_k\} \end{aligned}$$

On this directed graph we do a search for the shortest $s-t$−path which is possible in $O(n)$ since there are at most $2n$ unique edges in the graph. Calculating the edge costs is possible in $O(n)$ when going through the travel requests chronologically.
The shortest path now corresponds to the optimal decisions and its costs to the costs of the bought tickets.

Exercise 1.2

(a)

Proposition 1.1. *For each phase $[\tau_i, \tau_{i+1}$ with $1 \le i \le k$ the interval $[\tau_i, \tau_i + T$ is expensive.*

Proof. At τ_i OPT buys a BahnCard. Assume, $[\tau_i, \tau_i + T)$ is not expensive. Then it is cheaper not to buy a BahnCard than to buy one. Therefore OPT would not buy a BahnCard at τ_i. This contradicts the fact that OPT buys a BahnCard at τ_i. □

Proposition 1.2. *Any subinterval of $[\tau_i + T, \tau_{i+1})$ of length smaller than T is cheap.*

Proof. Assume $[\tau_i + T, \tau_{i+1})$ is expensive. Then it would be optimal to buy a BahnCard at the time of the next travel request t_j after $\tau_i + T$. Since OPT makes the optimal choice, it would buy a BahnCard at t_j. Since $t_j \notin \{\tau_1, \dots \tau_k\}$, OPTdoes not buy a BahnCard. Therefore $[\tau_i + T, \tau_{i+1})$ can't be expensive. □

(b)

Proposition 1.3. *For $I = [0, \tau_1)$, it holds that $c_{SUM} \le c_{OPT}$.*

Proof. Since OPT buys its first BahnCard at τ_1, $c_{OPT} = \sum_{i:t_i \in I} c_i$ and I can't be expensive, thus $\sum_{i:t_i \in I} c_i < c^*$. SUM buys a BahnCard as soon as for j the sum $\sum_{k:t_k \in [t_j - T, t_j)} c_k \geq c^*$. Since this can't be true at any time in I, SUM does not buy a BahnCard in I. Therefore

$$c_{SUM} = \sum_{i:t_i \in I} c_i = c_{OPT} \Rightarrow c_{SUM} \leq c_{OPT} \text{ for } I = [0, \tau_1).$$

□

(c)

If *SUM* doesn't buy a BahnCard, then $c_{SUM} = \sum_{j:t_j \in I} c_j$. For *OPT* the costs would be

$$c_{OPT} = C + \sum_{j:t_j \in [\tau_i, \tau_i + T)} \beta c_j + \sum_{j:j_j \in [\tau_i + T, \tau_{i+1})} c_j$$

The worst case for *SUM* would be that *OPT* only pays discounted prices and therefore

$$c_{OPT} = C + \sum_{j:t_j \in [\tau_i, \tau_i + T)} \beta c_j = C + \beta c_{SUM}$$

$$\begin{aligned}
c_{SUM} &\leq c_{SUM} + C + (1 - \beta)C \\
&= 2C + \beta \cdot c_{SUM} \\
&= C + \beta \cdot c_{SUM} + \beta C + (1 - \beta)C \\
&\leq C + \beta \cdot c_{SUM} + (1 - \beta)\beta \cdot c_{SUM} + (1 - \beta)C \\
&= c_{OPT} + (1 - \beta)c_{OPT} \\
&= (2 - \beta)c_{OPT}
\end{aligned}$$

(d)

We partition the interval I into three intervals I_1, I_2, I_3. I_1 is the timeframe in which *SUM* still owns a BahnCard that was bought in the time before τ_i. I_2 is the interval in which *SUM* doesn't own a BahnCard and I_3 the interval starting when *SUM* buys a new BahnCard. For each interval we denote the costs of the travel requests in it as:

$$C_j = \sum_{k:t_k \in I_j} c_k$$

Since *OPT* buys a BahnCard at τ_i that expires at $\tau_i + T$, the interval $[\tau_i, \tau_i + T)$ must be expensive. Since *SUM* buys a BahnCard whenever the current request makes the previous interval expensive, it buys a BahnCard at t' which is the time of the last travel request (t', c') before $\tau_i + T$. Therefore $I_3 = [t', \tau_i + 1)$

We can assume w.l.o.g. that there are no non-discounted requests for *SUM* in I_3, since both *SUM* and *OPT* would have to pay the full price. The worst case for *SUM* would be, if C_1 and C_3 would be very low and C_2 very high, since I_2 is the only interval in which only *OPT* gets the discount.

$$
\begin{aligned}
\frac{c_{SUM}}{c_{OPT}} &= \frac{C + \beta C_1 + C_2 + \beta C_3}{C + \beta C_1 + \beta C_2 + \beta c' + C_3 - c'} \\
&= \frac{C + \beta C_1 + C_2 + \beta C_3}{\frac{C}{1-\beta} + C_3 - c'} \\
&\leq \frac{C + C_1 + C_2 + C_3}{\frac{C}{1-\beta} + C_3 - c'} \quad | \text{ in the worst case, } SUM \text{ has to pay almost all the full prices} \\
&= \frac{C + C_2}{\frac{C}{1-\beta}} \quad | \text{ Let } C_1, C_3 \text{ be very small} \\
&= \frac{(1-\beta)(C + C_2)}{C} \\
&= \frac{(1-\beta)C}{C} + \frac{(1-\beta)C_2}{C} \\
&\leq 1 - \beta + 1 \quad | \, C \text{ must be smaller than the saved costs} \\
&= (2 - \beta)
\end{aligned}
$$

Exercise 1.3

(a)

Proposition 1.4. *A is c-competitive, i.e.* $\sum_{i=1}^{n} c_A(i) \leq c \cdot \sum_{i=1}^{n} c_{OPT}(i)$.

Proof.

$$
\begin{aligned}
\sum_{i=1}^{n} c_A(i) &\leq \sum_{i=1}^{n} c_A(i) + \sum_{i=1}^{n} \Phi_r(i) - \Phi_r(i-1) \mid \text{ since } \Phi_r \text{ can't be negative} \\
&= \sum_{i=1}^{n} c_A(i) + \Phi_r(i) - \Phi_r(i-1) \\
&\leq \sum_{i=1}^{n} c \cdot c_{OPT}(i) \\
&= c \cdot \sum_{i=1}^{n} c_{OPT}(i)
\end{aligned}
$$

□

(b)